11-DAY CODING BOOTCAMP

DAY 1

HTML FROM SCRATCH

Build your first webpage and publish it to GitHub

Today's big achievement
Create a personal profile webpage, understand the basic HTML structure, attempt five beginner logic problems, and upload the project to GitHub.

Designed for: First-year students who are absolute beginners.
Environment: VS Code + Browser
Session duration: 3 hours (180 minutes)

Learning approach: Explain simply → demonstrate → let students practise → build a small project → revise → publish.

1. Today's learning goals

By the end of this session, students should be able to:

• Explain what HTML is and why websites need structure.

• Explain the simple roles of HTML, CSS and JavaScript.

• Create an HTML file in VS Code and open it in a browser.

• Recognize tags, elements and attributes.

• Use headings, paragraphs, links, images and lists.

• Build a personal profile webpage and explain the important code.

• Attempt five beginner logic problems using step-by-step thinking.

• Create a GitHub repository and upload their project.

2. Three-hour session timetable

Time	Activity	Duration
0:00–0:10	Introduction and setup	10 min
0:10–0:40	HTML concepts from scratch	30 min
0:40–1:15	Live coding and HTML elements	35 min
1:15–1:50	Personal Profile mini project	35 min
1:50–2:20	Five logic-building problems	30 min
2:20–2:40	Git & GitHub practical	20 min
2:40–2:50	Exam preparation and quiz	10 min
2:50–3:00	Revision and buffer	10 min
TOTAL		**180 min**

Timing note: Keep the project small. If setup or GitHub authentication takes longer, protect understanding rather than rushing students. Use the final buffer to finish unfinished steps.

3. Start the class: connect coding to real life

Mentor says: “Good morning, everyone! Have you ever visited Instagram, Amazon or YouTube? Have you wondered how a website shows text, images, buttons and different sections? Today, we will create our first webpage.”

Ask the class: What do you see on a website? Can a website contain only text? How can we display an image? Can we create a website without writing code?

Show a simple website in the browser. Point to its heading, paragraph, image, link and button. Tell students that HTML gives webpage content its structure.

4. The big picture: HTML, CSS and JavaScript

Use the house analogy: HTML is like the walls, doors and rooms of a house. CSS is like paint and decoration. JavaScript is like switches and lights that make things happen.

Technology	Simple meaning	Example
HTML	Creates structure	A button exists on the page
CSS	Makes it look good	The button becomes blue
JavaScript	Adds interaction	Clicking the button shows a message

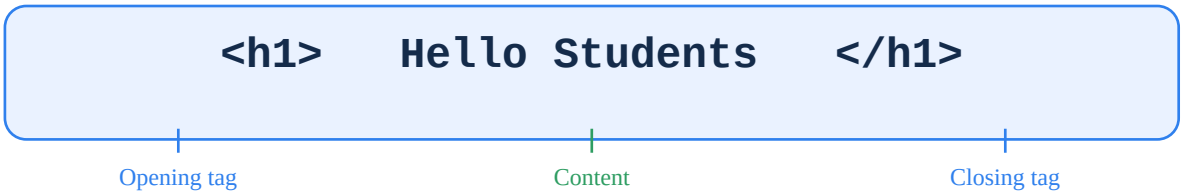
Ask: If we want to create a heading, which technology do we use? **Answer:** HTML.

5. What is HTML?

HTML stands for HyperText Markup Language. It is a markup language used to create the structure of webpages. HTML is not a programming language.

Explain the words simply:
• HyperText: text that can connect to other pages.
• Markup: special tags describe the content.
• Language: a way to give structured instructions to the browser.

6. What is a tag and what is an element?



In **<h1>Hello Students</h1>**, the first part is the opening tag, the words are the content, and the last part is the closing tag. Together, they make an HTML element.

7. Headings

```
<h1>Main Heading</h1>
<h2>Second Heading</h2>
<h3>Third Heading</h3>
<h4>Fourth Heading</h4>
<h5>Fifth Heading</h5>
<h6>Smallest Heading</h6>
```

Explain using a book: H1 is the book title, H2 is a chapter, and H3 is a topic inside a chapter. H1 is the highest heading level; H6 is the lowest.

8. Paragraphs

```
<p>My name is Rahul.</p>
<p>I am a first-year student.</p>
```

The **<p>** tag creates a paragraph. Each paragraph is a separate block of text.

9. Attributes, links, images and lists

An **attribute** gives extra information about an HTML element. Attributes are written inside the opening tag.

Example: link

```
<a href="https://google.com">Visit Google</a>
```

Explain: a is the anchor tag; `href` is the attribute; the URL is its value; “Visit Google” is the clickable text.

Example: image

```
<img src="https://placehold.co/200x150" alt="Sample image">
```

Explain: `img` displays an image; `src` tells the browser where to find it; `alt` describes the image if it cannot be displayed or for assistive technology.

Example: ordered list

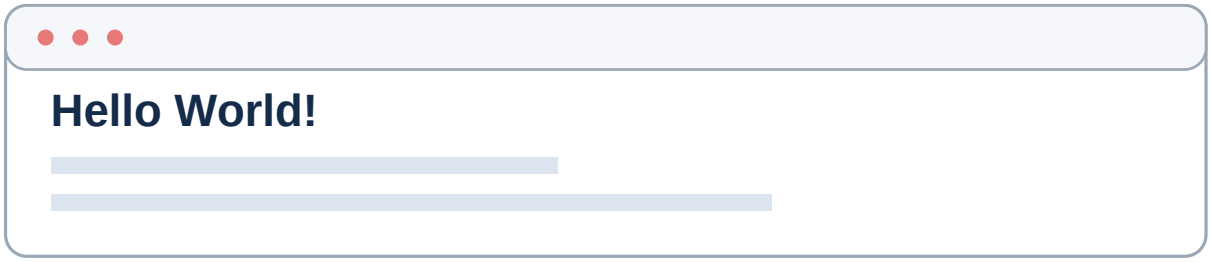
```
<h2>My Subjects</h2>
<ol>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
</ol>
```

Example: unordered list

```
<h2>My Hobbies</h2>
<ul>
    <li>Playing Cricket</li>
    <li>Reading</li>
    <li>Coding</li>
</ul>
```

Remember: OL = ordered list (numbered); UL = unordered list (bullets); LI = list item. The elements `img`, `br` and `hr` are void elements and do not need closing tags.

10. Live coding: create your first webpage



Step 1 — Create a project folder

1. Create a folder named **`day-1-html`**.
2. Open VS Code.
3. Choose File → Open Folder and select the folder.
4. Create a new file named **`index.html`**.

Step 2 — Type the basic HTML structure

```
<!DOCTYPE html>
<html>
<head>
    <title>My First Website</title>
</head>
<body>
    <h1>Hello World!</h1>
    <p>This is my first webpage.</p>
</body>
</html>
```

Code	Meaning
<!DOCTYPE html>	Tells the browser this is an HTML5 document.
<html>	The root element of the page.
<head>	Contains page information and metadata.
<title>	Sets the browser tab title.
<body>	Contains visible webpage content.
<h1>	Creates a main heading.
<p>	Creates a paragraph.

Step 3 — Run it: Save using Ctrl+S (Windows) or Cmd+S (Mac). Open the folder and double-click index.html. The browser displays “Hello World!”.

Ask: Can you see your first webpage? What changed when you edited the text and refreshed the browser?

11. Mini project: My Personal Profile Page

Hi, I am a student!

About me • Hobbies • Favorite subjects

My first HTML profile page

Project goal: Combine the HTML elements learned today to make a small introduction page. Students should personalize the name, introduction, hobbies and subjects.

Project requirements

• Student's name as the main heading.

• An “About Me” heading and introduction paragraph.

• A profile image with alt text.

• A list of hobbies.

• A list of favorite subjects.

• A link to a favorite website.

Suggested 35-minute project flow

5 min: explain requirements • 15 min: students build • 10 min: personalize • 5 min: check output and explain code.

Complete reference code

```
<!DOCTYPE html>
<html>
<head>
    <title>My Profile</title>
</head>
<body>
    <h1>Hi, I am Rahul!</h1>

    <h2>About Me</h2>
    <p>I am a first-year student.
       I love learning new things.
       I want to become a software developer.</p>

    <h2>My Profile Picture</h2>
    <img src="https://placehold.co/200x200"
         alt="My profile picture">

    <h2>My Hobbies</h2>
    <ul>
        <li>Playing Cricket</li>
        <li>Listening to Music</li>
        <li>Coding</li>
    </ul>

    <h2>My Favorite Subjects</h2>
    <ol>
        <li>Mathematics</li>
        <li>Computer Science</li>
        <li>English</li>
    </ol>

    <h2>My Favorite Website</h2>
    <a href="https://youtube.com">Visit YouTube</a>
</body>
</html>
```

Mentor check: Ask two students to share their screens and explain why they used H1, P, SRC, ALT, OL and UL. Focus on structure, not visual beauty. Styling comes in the CSS sessions.

12. Five logic-building problems

Because students are absolute beginners, Day 1 focuses on logical thinking and pseudocode. From Day 4, the same ideas will be implemented in JavaScript.

1. Add two numbers

Input: 10 and 20 | Output: 30

```
START
A = 10
B = 20
SUM = A + B
PRINT SUM
END
```

Explain: Take two values, add them, and display the result.

2. Even or odd

Input: 8 | Output: Even

```
START
NUMBER = 8
IF NUMBER % 2 == 0
    PRINT "Even"
ELSE
    PRINT "Odd"
END
```

Explain: A number is even when dividing it by 2 leaves remainder 0. The % symbol means remainder.

3. Largest of two numbers

Input: 15 and 25 | Output: 25

```
START
A = 15
B = 25
IF A > B
    PRINT A
ELSE IF B > A
    PRINT B
ELSE
    PRINT "Both are equal"
END
```

Explain: Compare the two values. Also handle the case where both are equal.

4. Positive, negative or zero

Input: -5 | Output: Negative

```
START
NUMBER = -5
IF NUMBER > 0
    PRINT "Positive"
ELSE IF NUMBER < 0
    PRINT "Negative"
ELSE
    PRINT "Zero"
END
```

Explain: Check greater than zero first, less than zero next, otherwise it is zero.

5. Print numbers from 1 to 10

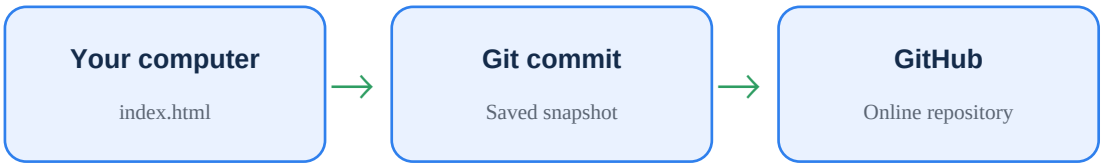
Output: 1 2 3 4 5 6 7 8 9 10

```
START
NUMBER = 1
WHILE NUMBER <= 10
    PRINT NUMBER
    NUMBER = NUMBER + 1
END WHILE
END
```

Explain: Start at 1, print it, add 1, and repeat while the number is at most 10.

Practice rule: Do not show the answer immediately. Give students time to read, think, explain their steps in simple English and then write pseudocode.

13. Git & GitHub: publish the project



Git tracks changes in code. GitHub hosts repositories online so developers can store and share code.

Step 1 — Create a GitHub repository

1. Visit github.com and sign in or create an account.
2. Click the + icon → New repository.
3. Name it **`day-1-html-profile`**.
4. Choose Public (if appropriate for the class) and create the repository.

Step 2 — Open VS Code terminal

```
git --version
```

This checks whether Git is installed. If not, install it from git-scm.com/downloads.

Step 3 — Run these commands one at a time inside the project folder

```
git init
git add .
git commit -m "Created my first HTML profile"
git branch -M main
git remote add origin YOUR_REPOSITORY_URL
git push -u origin main
```

What the commands mean

- `git init`: start tracking this project.
- `git add .`: prepare all project files for the next commit.
- `git commit`: save a snapshot with a message.
- `git branch -M main`: name the current branch main.
- `git remote add origin`: connect the local project to the GitHub repository.
- `git push`: upload the commit to GitHub.

Final check: Refresh the GitHub repository page. Students should see index.html. If authentication or installation takes longer, finish the guided setup without rushing the learning.

14. Exam preparation: theory and practical

Q1. What is HTML?

HTML stands for HyperText Markup Language. It is used to create the structure of webpages.

Q2. Is HTML a programming language?

No. HTML is a markup language.

Q3. What is an HTML tag?

A markup instruction written inside angle brackets, such as .

Q4. What is an HTML element?

An element generally consists of an opening tag, content and a closing tag.

Q5. What is an attribute?

An attribute provides additional information about an HTML element.

Q6. Difference between ol and ul?

OL creates an ordered list; UL creates an unordered list.

Q7. What does href do?

It specifies the destination of a link.

Q8. What does alt do?

It provides alternative text describing an image.

Practical exam practice

• Create a webpage with your name as the main heading.

• Add two paragraphs.

• Create an ordered list of five subjects.

• Add an image with alternative text.

• Create a link to Google.

15. Final class checklist

• Every student created index.html.

• Every student understood the HTML document structure.

• Every student used headings and paragraphs.

• Every student created a link and image.

• Every student created ordered and unordered lists.

• Every student completed the Personal Profile project.

• Every student attempted all five logic problems.

• Every student created or prepared a GitHub repository.

• Every student attempted the knowledge check.

16. Homework (15–20 minutes, optional)

• Change the content of the profile page.

• Add three more hobbies.

• Add a link to another website.

• Explain the difference between an HTML tag, element and attribute in your own words.

Day 1 success measure

A student has succeeded today if they can independently create and open a basic HTML page, explain its important elements, attempt the five logic questions, and save their project locally or on GitHub.

Next session: Day 2 — CSS from scratch. We will connect CSS to HTML and learn selectors, colors, fonts, backgrounds, borders, margin, padding and the box model. The achievement will be a styled personal profile card.